INCH-POUND
MIL-DTL-24643/41C
22 August 2002
SUPERSEDING
MIL-C-24643/41B
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SAU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First -	Copper conductor, uncoated, ASTM B 286, AWG Size 20, 7 strands.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Crosslinked polyethylene insulation. Minimum wall thickness 0.015 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
Fourth -	Braided shield of AWG No. 34 or No. 36 uncoated copper. Braid angle of 30 to 35 degrees minimum coverage of 85 percent.
Fifth -	Shield insulation of two polyester tapes, per NEMA FI 4, outer tape sealed. An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

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Construction (watertight) (continued)

Sixth - The required number of singles (see table I) cabled together with a lay of not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward.
Seventh - Binder tape, applied helically with overlap.
Eighth - Cross-linked polyolefin jacket. Minimum average thickness 0.063 inch.

TABLE I. Details.

Military part no. M24643/41	Size	Number of conductors	Overall diameter (inches)	
			TYPE LS1SAU	
			min	max
-01UN	44	44	0.910	0.990

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor resistance - ohms/1000 feet at 25°C, maximum	10.71
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	2000
Shield to shield	500
Insulation resistance - Megohms -1000 feet, minimum	
Conductor to shield	500
Component shield to shield	100
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Capacitance - 1 kilohertz (kHz), pF/ft, maximum	72
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160

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Group C:
Physicals (aged)
- Insulation
  - Air oven
    - Tensile strength - percent of unaged, minimum 80
    - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged, minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (jacket) - cycles, minimum 125
- Cable sealant removability ..... No failure
- Shield - conformance to material, construction and coverage ..... No failure

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable)(125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45

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Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum.....80
Weathering (jacket)..... No failure

UNIT ORDERING LENGTHS: 1000 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-039)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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